

## **Description of Additional Supplementary Files**

**File Name: Supplementary Data 1**

**Description:** Selected Data Fields from the UKB dataset, categorized into essential information, detailed information, and minor information.

**File Name: Supplementary Data 2**

**Description:** Missing pattern analysis for different fields in the UKB dataset.

**File Name: Supplementary Data 3**

**Description:** Data dictionary for the selected variables in the All of Us dataset.

**File Name: Supplementary Data 4**

**Description:** ICD-to-Phecode correspondence for multi-disease prediction and risk assessment, comparing our method with other approaches for the same disease definitions.